CS 215: Task 07: Due: 15-Oct; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix of the textbook (5th edition).

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. Fill volumes of a bottling machine are $\mathcal{N}(\mu, \sigma^2)$ with $\sigma = 4$ ml. A random sample of $n = 16$ bottles gives $\bar{x} = 502$ ml.
   (a) If samples of 16 bottles were repeatedly drawn in this way, find an interval around the observed sample mean 502 ml such that the corresponding interval would contain μ in approximately 95% of repeated samples.

   (b) With the conditions as in part (a), find the smallest sample size for which the margin of error (half-width of the interval) is at most 1 ml.

   (c) How many bottles should we sample if we want the error to be at most 0.5 ml instead?

(d) Suppose the interval obtained in part (a) is written as

$$502 \pm \delta.$$

If a random sample of 100 individual bottles is taken, would you expect about 95 of them to have fill volumes within

$$502 \pm \delta?$$

Explain your answer.

2. Your supervisor asks you to estimate the mean running time of a piece of code. You run the program independently, and obtain the following running times (in ms):

$$12,\ 15,\ 11,\ 14,\ 13,\ 16,\ 10,\ 14,\ 12.$$

Assume that the running times are normally distributed.

(a) How would you report the mean running time to your supervisor? Use a 95% confidence level.

(b) Your assistant wrongly uses 1.96 in place of the t critical value. Compute the assistant's reported interval and explain the consequence for its actual coverage.

(c) A second assistant constructed a 99% confidence interval. Compare its width to the one in part (a).

3. Let $Z \sim \mathcal{N}(0,1)$.

(a) Show that

$$M_{Z^2}(t) = E\left[e^{tZ^2}\right] = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.$$

(b) Let $Z_1, \ldots, Z_k$ be i.i.d. $\mathcal{N}(0,1)$ and $W = \sum_{i=1}^{k} Z_i^2$. Find $M_W(t)$.

(c) Using $M_W(t)$, show that $E[W] = k$ and $\text{Var}(W) = 2k$.

(d) If $W_1 \sim \chi^2_a$ and $W_2 \sim \chi^2_b$ are independent, what is the distribution of $W_1 + W_2$? Justify your answer.

4. Let $X_1, \ldots, X_{10}$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and let S^2 be the sample variance.

   (a) Find $E[S^2]$, $\text{Var}(S^2)$ and the standard deviation of S^2 expressed as a multiple of σ^2.

(b) Find constants c_1 and c_2 such that

$$P(S^2 < c_1\sigma^2) = 0.05, \qquad P(S^2 > c_2\sigma^2) = 0.05.$$

5. A random sample of $n = 11$ resistors from a normal population gives the sample standard deviation $s = 0.8\ \Omega$.

(a) Construct a 95% confidence interval for σ^2.

(b) Construct a 95% confidence interval for σ.

(c) Explain why the interval in part (a) is not symmetric about s^2.

(d) The specification requires $\sigma \le 1.5\ \Omega$. Based on your interval, what can you conclude at the 95% confidence level?

6. Let $X_1, \dots, X_5$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ and define

$$Y = c\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.$$

(a) Find the constant $c > 0$ for which Y has a student's t distribution, and state its degrees of freedom. c should not depend on σ.

(b) State which independence property you used in your construction.